

# Assignment 2: Design Your Dream Logo

Logo Name: **Nimbus**

Aishwarya J S

August 6, 2026

## Concept

**Nimbus** is an imaginary weather-forecasting app. The logo shows a sun rising behind a three-puff cloud, sitting above a rolling green horizon with two small mirrored hills and gentle wind lines. The whole design is built around a single **vertical mirror axis** ( $x = 0$ ): every element on the right has a matching mirrored twin on the left.

## Equations Used (15 total)

| #  | Equation (Desmos syntax)                             | Type                | Represents                         |
|----|------------------------------------------------------|---------------------|------------------------------------|
| 1  | $x^2 + (y - 5)^2 = 2.25$                             | Circle              | Sun boundary                       |
| 2  | $x^2 + (y - 5)^2 \leq 2.25$                          | Circle (inequality) | Sun fill (color)                   |
| 3  | $x = 0, \{6.5 \leq y \leq 7.2\}$                     | Linear              | Sun ray, top                       |
| 4  | $y = x + 5, \{1.6 \leq x \leq 2.1\}$                 | Linear              | Sun ray, upper-right               |
| 5  | $y = -x + 5, \{-2.1 \leq x \leq -1.6\}$              | Linear              | Sun ray, upper-left (mirror of #4) |
| 6  | $(x + 1.5)^2 + (y - 1)^2 = 0.81$                     | Circle              | Cloud puff, left                   |
| 7  | $x^2 + (y - 1.3)^2 = 1.69$                           | Circle              | Cloud puff, center                 |
| 8  | $(x - 1.5)^2 + (y - 1)^2 = 0.81$                     | Circle              | Cloud puff, right (mirror of #6)   |
| 9  | $y = 0.3, \{-2.4 \leq x \leq 2.4\}$                  | Linear              | Cloud base line                    |
| 10 | $x^2 + (y - 1.3)^2 \leq 1.69 \{y \geq 0.3\}$         | Circle (inequality) | Cloud fill (color)                 |
| 11 | $y = 0.15x^2 - 3.5, \{-3.2 \leq x \leq 3.2\}$        | Quadratic           | Rolling horizon                    |
| 12 | $y = -0.6 x + 1.8  - 2.5, \{-2.7 \leq x \leq -0.9\}$ | Absolute value      | Hill, left                         |
| 13 | $y = -0.6 x - 1.8  - 2.5, \{0.9 \leq x \leq 2.7\}$   | Absolute value      | Hill, right (mirror of #12)        |
| 14 | $y = 0.12 \sin(4x) - 1.7, \{-2.6 \leq x \leq -0.4\}$ | Trigonometric       | Wind gust, left                    |
| 15 | $y = 0.12 \sin(4x) - 1.7, \{0.4 \leq x \leq 2.6\}$   | Trigonometric       | Wind gust, right (mirror of #14)   |

Five function types appear: **linear, quadratic, circular, absolute value, and trigonometric**. Inequalities (#2, #10) shade the sun and cloud with color instead of leaving them as outlines. Symmetry is clearly visible: equations #4/#5, #6/#8, #12/#13, and #14/#15 are each mirror pairs reflected across the vertical axis  $x = 0$ , so the logo looks identical if flipped left to right.

## Mathematics Behind the Design

The logo relies on reflective symmetry rather than rotational symmetry, which is achieved simply by pairing every off-center equation with a version whose horizontal shift has the opposite sign –

for instance  $(x - 1.5)^2 + (y - 1)^2 = 0.81$  mirrored to  $(x + 1.5)^2 + (y - 1)^2 = 0.81$  for the cloud puffs, and  $|x - 1.8|$  mirrored to  $|x + 1.8|$  for the hills. The three cloud circles overlap so their union reads as a single soft cloud shape, while the inequality  $x^2 + (y - 1.3)^2 \leq 1.69$  restricted to  $y \geq 0.3$  fills only the puff above the cloud's flat base, avoiding an ugly circular bottom edge. The quadratic horizon curves gently upward on both sides since its coefficient is positive, giving a natural "bowl" shape for the ground, and the two absolute value hills sitting on it use  $|x - c|$  shifts to place identical V-shaped peaks symmetrically without duplicating the underlying math. The sine-based wind lines were chosen over straight lines specifically to suggest motion and air, something no polynomial or absolute value function can convey as naturally.

## Preview (TikZ reconstruction of the Desmos graph)

